

**M A S A R Y K
U N I V E R S I T Y**

FACULTY OF INFORMATICS

**Musikk: A Modern, Social-First
Music Streaming Platform**

Bachelor's Thesis

KIRILL VOROZHTSOV

Brno, Fall 2025

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Advisor: Mgr. Luděk Bártek, Ph.D

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Declaration

Hereby I declare that this paper is my original authorial work, which I have worked out on my own. All sources, references, and literature used or excerpted during elaboration of this work are properly cited and listed in complete reference to the due source.

During the preparation of this thesis, the following AI tools were used:

- **ChatGPT:** Used for debugging and making small code corrections, as well as LaTeX formatting.
- **Grammarly:** Used for text correction.

Kirill Vorozhtsov

Advisor: Mgr. Luděk Bártek, Ph.D

Acknowledgements

I would like to thank Mgr. Luděk Bártek, Ph.D., for his time and the valuable advice he provided for this thesis.

Abstract

This bachelor's thesis aims to design and implement a web-based music streaming platform focused on social interactions between users.

The theoretical part presents a survey that analyzes people's music listening habits, showing that such a platform could find its user base. It then compares existing music-related services with built-in social features and derives functional and non-functional requirements for the proposed system. Based on these requirements, suitable technologies and architectural patterns are selected.

The practical part covers the technical aspects of the platform, including data model, audio processing, user interaction features, and real-time synchronization, among others.

The result is a fully functional and deployed application.

Keywords

Audio Streaming, Python, JavaScript, MPEG-DASH, HLS, WebSockets

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1 Introduction

Music streaming has been significantly growing in popularity each year for the past decade[1]. And this is to be expected; numerous studies showed how important music is for people. For instance, Rentfrow in his 2012 paper[2] shows that it can influence mood, change self-perception, and help with social bonding. In addition, the rise of the clubbing scene, music festivals, and other social music-related activities clearly indicates the important social role of music[3].

Despite this, it is surprising that features that facilitate social interactions are not widely implemented in the existing music streaming platforms, as will be shown in 3.

The goal of this thesis is to design a music-centric platform that provides capabilities for collaboration and social interaction around music. This work is divided into the following **six** chapters:

- **Chapter 2** Presents the outcomes of a survey illustrating how people consume music, how prevalent it is in social interactions, and why this type of platform can be relevant.
- **Chapter 3** Explores what existing streaming platforms and other music-related services already offer.
- **Chapter ??** Outlines the functional and non-functional criteria for the application.
- **Chapter ??** Describes the choices of technologies that are used by the application.
- **Chapter ??** Explains the development process and implementation details.
- **Chapter ??** Summarizes the results of the work.

2 Music Consumption Survey

To better support the choice of topic and demonstrate that a music streaming platform centered on social interactions could be a relevant service, a brief survey was conducted. It examined individual music listening and discovery habits, platform usage patterns, music-related interactions between people and the attitude towards possible social features on the platform.

2.1 Methods

The service chosen for the questionnaire was ‘Google Forms’[4], as it provides a simple interface for survey creation, allows for easy sharing of the form, and supports result extraction as a ‘csv’ file.

The questionnaire itself consisted of 15 questions. Most of them are multiple-choice and closed-ended, with some allowing a custom answer. Custom answers are grouped under the “other” answer in the provided figures. Answers in their original form can be found in the full survey.

The complete questionnaire is provided in Appendix A.1; an online version is also available.¹

2.2 Results

This section presents the survey outcomes and analyzes their significance. Not everything is discussed; for example, questions dedicated to music discovery are omitted.

A total of 119 people participated, with the majority being from Russian-speaking countries. The majority were in the 18 to 30 age group, with the exact distribution shown in Figure 2.1.

1. https://docs.google.com/forms/d/1vhhAu_SfuHV4xy6JoDaRsM5uCE1Yn_csTmN8u4Z1pHc

How old are you?

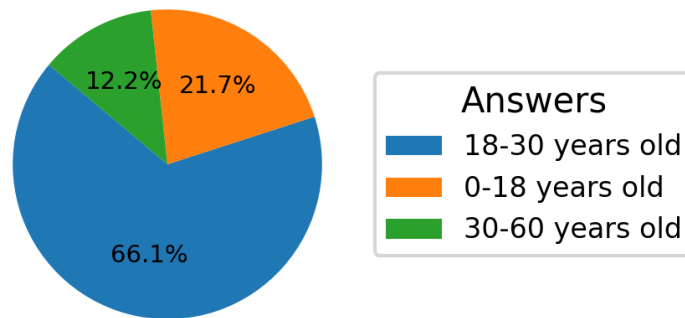


Figure 2.1: Age distribution of respondents

2.2.1 Engagement

Firstly, it was necessary to determine whether people actually do listen to music frequently and whether they use streaming platforms to do so.

Over 50% of respondents reported listening to more than 500 songs per month (i.e., 16 songs a day on average) (Figure 2.2), and nearly 80% stated that they listen to music daily (Figure 2.3).

The numbers clearly indicate that people have a significant interest in music, and the following logical step was to determine how the audio content is mostly accessed.

2.2.2 Listening Methods

As can be seen in Figure 2.4, the percentage of streaming platform usage across all age groups is nearing 100%, with slightly higher numbers in the younger age groups. Although downloaded files and physical media have their presence, especially among aged 30-60, they are usually only an auxiliary option next to streaming solutions.

The question dedicated to the popularity of individual platforms has shown that Spotify is the most used one, which aligns with global statistics[5]. However, Yandex Music, being in second place, deserves an explanation. As mentioned previously, most of the respondents

2. MUSIC CONSUMPTION SURVEY

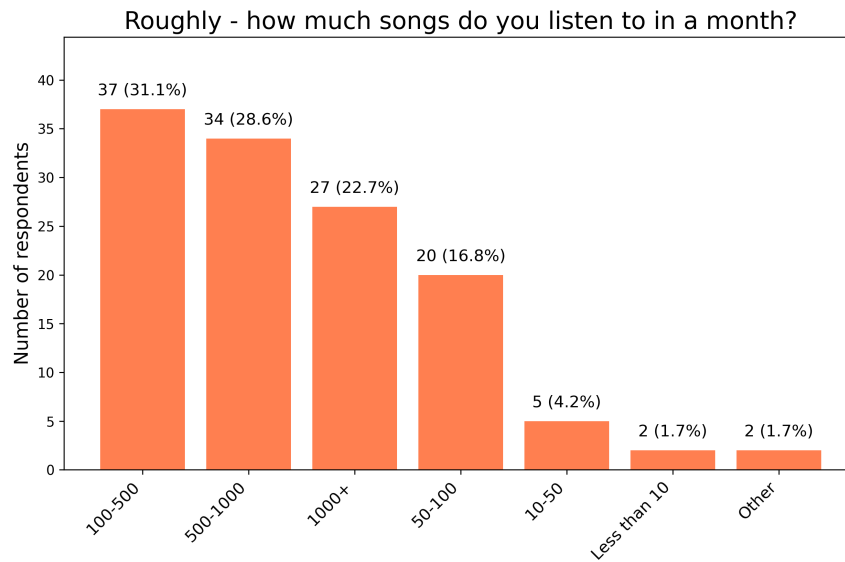


Figure 2.2: Monthly song count per respondent

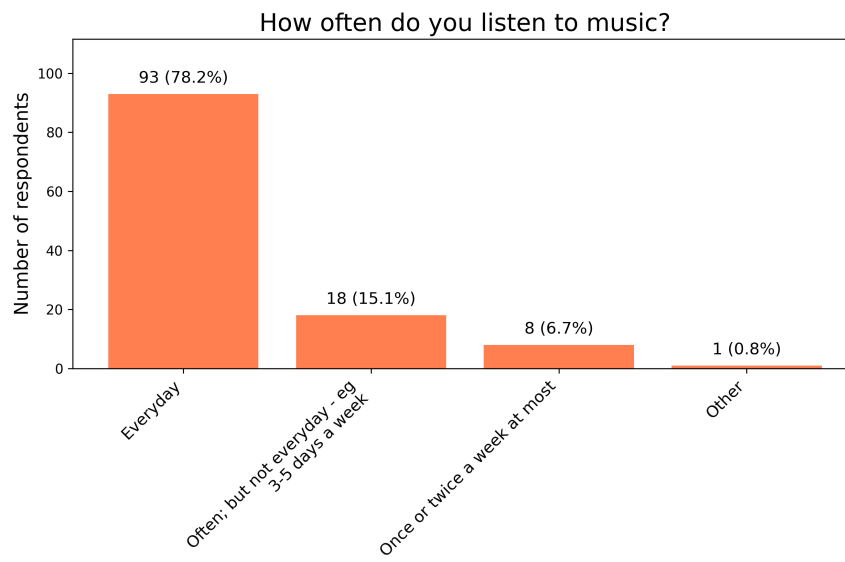


Figure 2.3: Listening frequency of respondents

2. MUSIC CONSUMPTION SURVEY

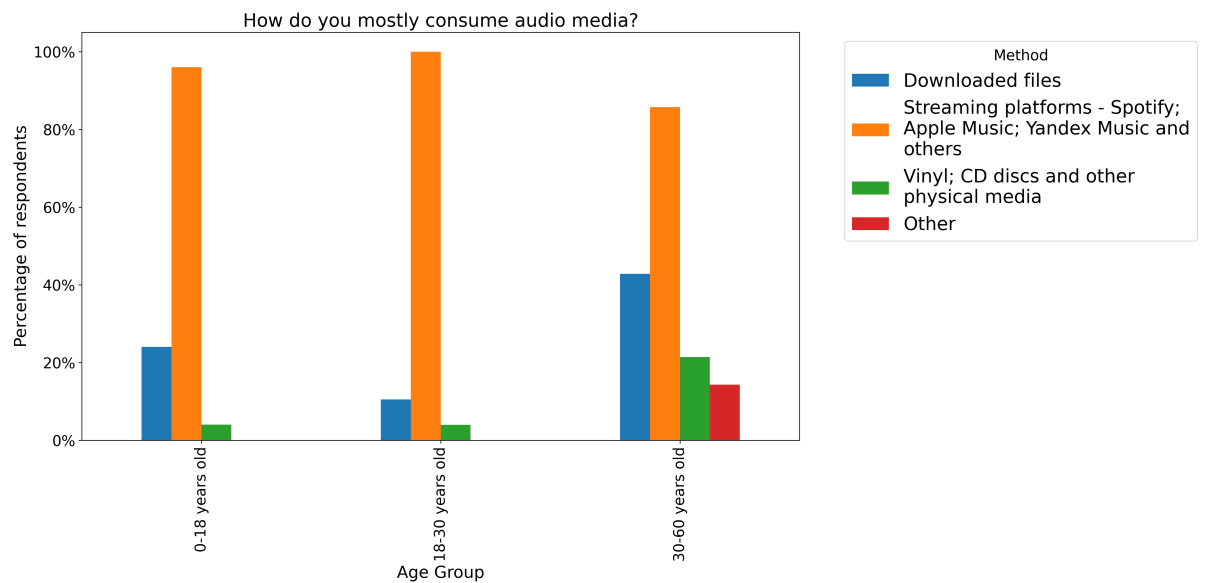


Figure 2.4: Audio media consumption by age group

are from Russian-speaking countries, specifically Russia. With many Western companies leaving its market in 2022, including music streaming services, most users have moved to locally available products: Yandex Music, VK Music, Zvuk, and others. Another notable point is the absence of other popular region-specific platforms, such as Amazon Music for the US market, QQ Music for the Chinese market, and JioSaavn for the Indian market, as all of the respondents were based in the European part of the world. The complete statistics are shown in Figure 2.5.

The survey showed that streaming services are indeed widely used. The next important step was to analyze the regularity of music-centered social interactions.

2.2.3 Social Interactions

The results of Question 10 indicate that nearly all respondents (99.2%) reported listening to music with others, primarily in offline settings such as gatherings or car rides (Figure 2.6). Online collaborative listening methods, although less common, were also mentioned by a

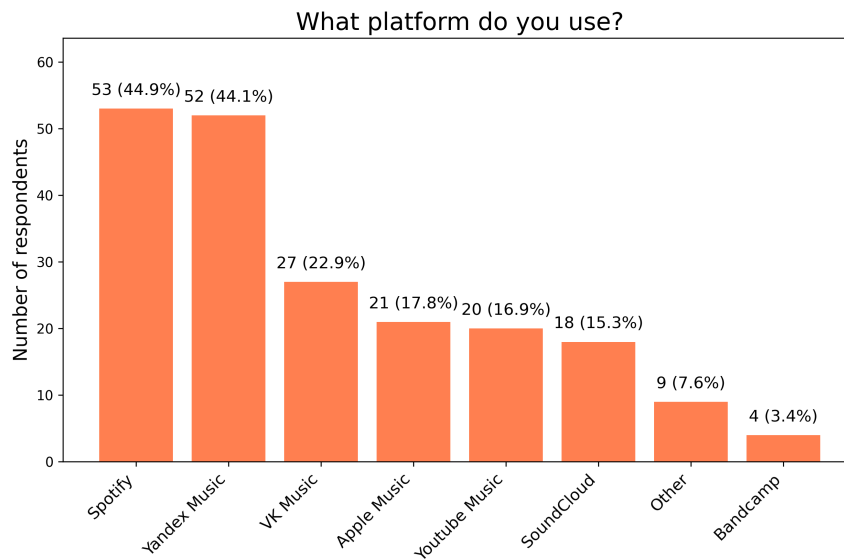


Figure 2.5: Streaming platforms popularity

quarter of respondents, indicating that digital solutions for shared listening are used, albeit not as prevalently as in-person scenarios.

Question 11 (Figure 2.7) examines the frequency of these interactions: the results show that a significant portion of respondents engage in shared listening regularly. Around 60% reported doing so at least a few times per month, with a smaller group indicating almost daily interactions. This supports the previously mentioned idea in Chapter 1 that music consumption is a common social activity.

The next question is phrased as follows: “How often do you/your friends share/discuss music?”. While the previous question focused on real-time in-person collective listening, this one highlights more intentional and in-depth musical interactions, such as sharing songs, recommending music, or discussing it. Music, in that case, is not just the background but the main purpose of engagement. The same trend as before can be observed in the responses: over 70% (Figure 2.8) indicated that they share music at least several times a month, with many doing so weekly or more often. This further emphasizes the active social role of music.

2. MUSIC CONSUMPTION SURVEY

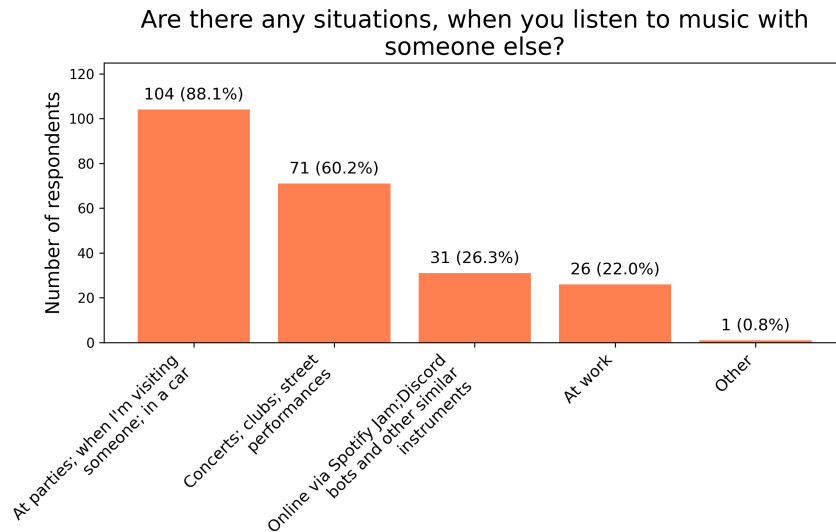


Figure 2.6: Social interaction methods

How often do you listen to music with someone else?

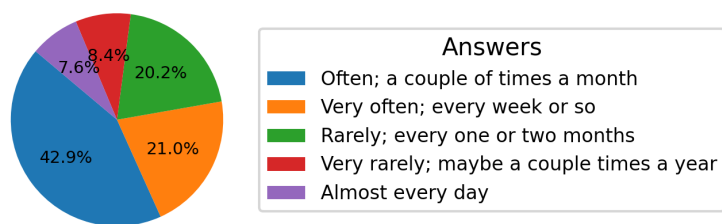


Figure 2.7: Social interactions frequency

How often do you/your friends share/discuss music?

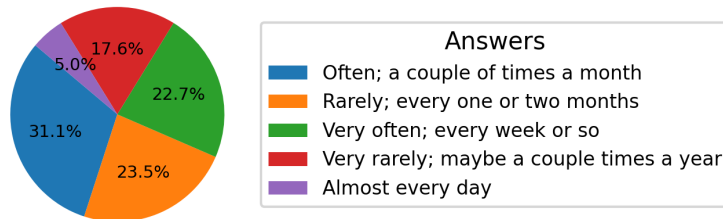


Figure 2.8: Music sharing frequency

Question 13 — “How does that happen?” (as in music sharing) — provides an insight into the specific means of sharing. Many respondents (Figure 2.9) stated that they typically send links from streaming platforms through external messaging apps. While this shows an apparent demand for sharing music online, it also reveals a gap: these interactions are fragmented across platforms. Hence, enabling such exchanges directly within the streaming app could provide a smoother and more uniform experience.

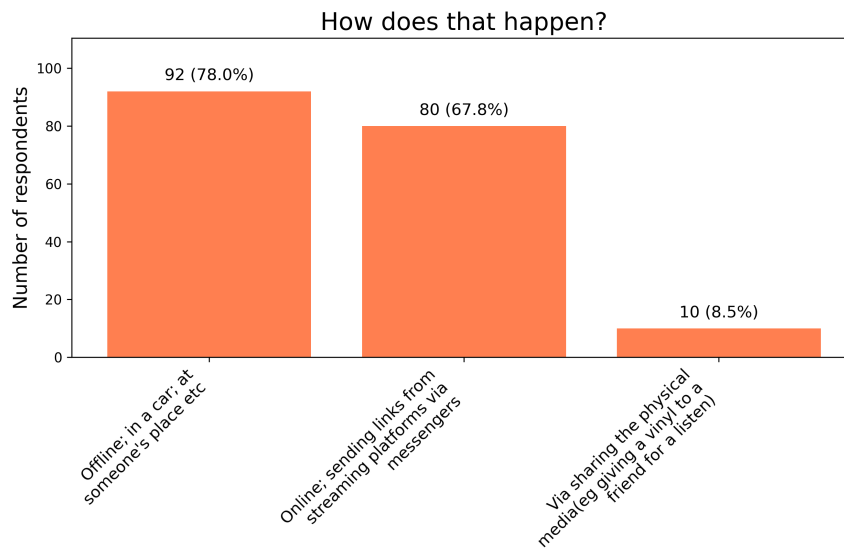


Figure 2.9: Music sharing methods

Responses to the question *“Would you say that you know a lot of people with similar music taste?”* were mixed, with a near-even split. This suggests that while some users already have a social circle with similar music taste, many do not. In addition, when asked *“Would you like to connect with others who share your musical taste, or influence those around you to explore and appreciate the music you enjoy?”*, the majority answered positively.

This data further supports the idea that social discovery features could be useful within a streaming platform. Moreover, over 60% of participants stated in Question 15 that they would use additional social features if available on a streaming platform.

2.3 Summary

Overall, the results show that people listen to music frequently and that streaming services are the primary medium for doing so. It is also evident that music is deeply involved in people’s lives. Moreover, it often appears in social interactions.

Yet, current platforms only partly support interactions with other users. Therefore, the idea of a music streaming service with integrated social features could be relevant, addressing real user needs.

3 Existing Platforms

In this chapter, existing platforms and services are examined to gain a better understanding of the instruments people use when interacting with music. Additionally, this research provides insight into the possible features to be implemented in the resulting application.

Firstly, streaming services are discussed, as the main aim of the thesis is to create a system for music streaming. Then, other music-related (e.g., forums) services are listed, as they offer alternative possibilities to engage with audio content, which can also be beneficial for the future platform. Lastly, platforms such as YouTube and TikTok are discussed, as the survey has shown that they are a popular way to interact with music 3.1.

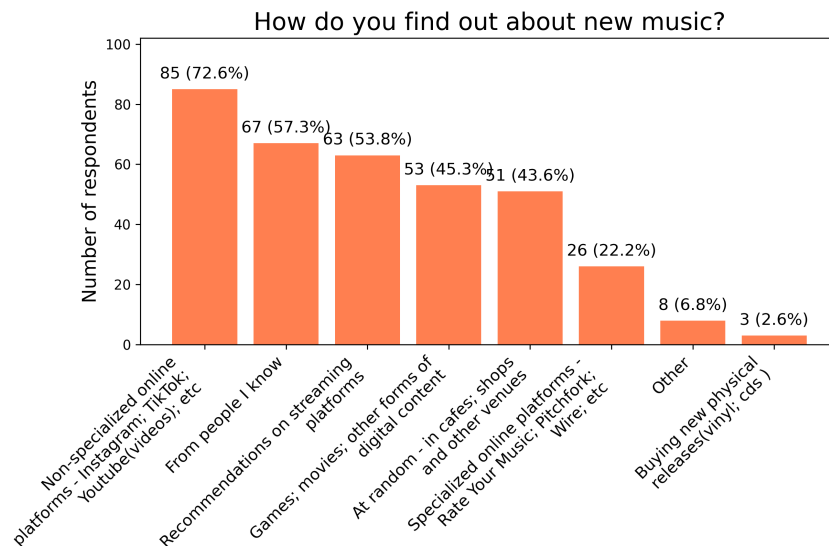


Figure 3.1: Streaming platforms popularity

3.1 Streaming Services

As can be seen in Figure 2.4 and further confirmed by the recent study of the International Federation of the Phonographic Industry[6], nowa-

days, the most common way of music consumption and discovery is through streaming services. Many options are available, but only those that are both popular and have unique elements will be considered in this section. The descriptions, rather than offering a general portrayal of each platform, will focus on the service-specific social features.

3.1.1 Spotify

One of the most prominent social features on ‘Spotify’ is the “Spotify Jam”[7]. It lets people create a collective song queue that is then synchronized among all connected users. Moreover, volume, the order of songs, and other playback aspects can be controlled individually by each user.

Another notable feature is “Friend Activity”[8], which shows what the people the user follows are currently listening to.

“Listening Parties” feature also deserves a mention: those are live chats with a limited capacity, that can be joined for a short time when new music is being released[9, 10]. This somewhat brings the offline collective music discovery experience to the online world.

During the implementation of this thesis (the text was initially written in Spring-Summer 2025) ‘Spotify’ has also added support for user-to-user chats (from September 2025) [11]. This further supports the claim that social features are in demand on music platforms.

3.1.2 VK Music

‘VK Music’ is a music service integrated into the ‘VK’ social network. Consequently, it is possible to send songs and playlists via private messages and add audio materials to posts in groups. ‘VK Music’ also provides an “Updates” tab, which is populated by the audio materials that any followed user or group has added to their “Liked Feed”[12].

3.1.3 SoundCloud

SoundCloud is one of the few platforms that lets its users leave comments and reactions on songs and playlists[13, 14]. In addition, each user has a feed consisting of personal uploads and reposts of other’s content[15], which is visible when visiting their profile.

3.1.4 Bandcamp

Every 'Bandcamp' user has a profile with four tabs: "collection", "wishlist", "followers", and "following". The most interesting feature is under the 'following' tab: users can subscribe to available genres and then specify the ones they want to showcase to others viewing their profile. The "wishlist" tab is also noteworthy, as 'Bandcamp's model blends streaming with the traditional purchases of individual releases, and in this tab, the users can show what they are looking forward to listening to in the future.

This concludes the streaming services section. Services that are focused on music, but not on the streaming itself, are discussed next.

3.2 Forums, Blogs, and other Music Media

Different resources are gathered under these umbrella terms, from professional music review websites to amateur and personal pages. Again, only widely used services that offer something noteworthy are listed.

3.2.1 Rate Your Music

'Rate Your Music is one of the largest music databases online. It is an incredible tool that can help you find and learn about new music to listen to.'[16]

This website is one of the biggest platforms with user-created content. Every member can write reviews and set ratings for music releases, contribute to the extensive "wiki" consisting of genres, thematic lists and charts, and connect to other members of the forum. However, the content is still moderated – any post has to be properly formatted and abide by the guidelines in order to appear on the website.

3.2.2 2step.ru

'2step.ru' is one of the better representatives of "old school" forums that is still active. This is a country-specific resource, and as a consequence, in addition to providing the regular forum and article sections,

it includes elements that are usually missing from larger portals. For example, a section about upcoming parties, a page dedicated to local and upcoming DJs, and an ongoing list of recorded radio shows[17].

3.2.3 last.fm

‘Last.fm’ is a music discovery and social networking service. Notable social features include friend lists, public groups for discussion, and the ability to comment on artist and track pages[18]. Additionally, an “Events” section aggregates concert listings and allows users to indicate that they are interested or confirm that they are going and to add comments under each event post, bridging online listening activity with real-world music events.

This section has introduced a couple of new possible features not previously seen in the listed streaming services. The next section examines platforms that are not primarily focused on music but influence how people engage with it.

3.3 Other Platforms

As mentioned earlier, one of the most interesting results of the survey is the high number of people discovering music on platforms that are not focused on music. Services like ‘Instagram’, ‘YouTube’, and ‘TikTok’, which are mainly used for sharing videos and other user content, have also become closely connected to how people find and listen to music. In recent years, these platforms have started to promote music more directly by making it possible to embed tracks into their platform-native content.

‘Instagram’, for example, added a special audio tool that lets users include music in their short videos (“Reels”[19]), helping songs reach more people through visual content.

‘YouTube’ automatically identifies songs used in videos and displays them in the description, making it easier to find and listen to them.

‘TikTok’ has had an even bigger impact – many songs have gone viral and later became hits in charts and streaming apps because

people use them in popular videos (these are often called “TikTok Hits”).

This shows that big platforms are not just reacting to users’ interest in music – they actively help promote it by connecting songs with content that people want to watch and share.

An integration with such services can be a big plus for the platform. A possible feature could be a badge saying “This track was in a TikTok user Alice123 sent you today!”.

Summary

The table 3.1 summarizes the comparison between the features mentioned in the sections above.

Difficulty suggests the possible complexity of implementation.

Mode assumes the typical context in which the feature operates.

Social Interaction Level indicates user involvement when using the feature.

Table 3.1: Comparison of Service-Specific Social Features

Feature	Interaction Level	Mode	Difficulty
Chat	Very High	Pair	High
Spotify Jam	High	Group	Very High
Forums/Wiki	High	Group	Medium
Personal/Friends Feed	High	Individual	Low
Song/Playlist Comments	High	Group	Low
Spotify Listening Parties	Medium	Group	Very High
Embedding Music Into Posts	Medium	Group	Medium
Friends’ Listening Activity	Medium	Pair	Low
Following Genres Display	Low	Individual	Medium
Real Life Events Section	Low	Group	High
Music Wishlist	Low	Individual	Low
External Services Integration	Low	Pair/Group	High

Overall, it is clear that numerous music-related instruments with social features built in are available across the Internet. However, services that focus on the actual audio delivery often support just a subset of the possible features. This, in turn, forces users to use multiple platforms in order to meet their needs. This project aims to bridge that gap, particularly by enhancing users' potential for social interaction.

Based on the research done in the previous and current chapters, these social features were chosen for implementation, as they provide a good balance between the complexity of implementation and the level of social interaction:

- Chat.
- Personal/Friends Feeds.
- Song/Playlist Comments.
- Friends' Listening Activity.

These and further requirements for the application are listed in the following chapter - ??.

A An appendix

A.1 Survey Questionnaire

Music Consumption Study

This short form is designed to analyze the people's behaviour related to music - ways of discovering new audio pieces, the role of music in social interactions, preferred ways of consuming audio information etc.

1. How old are you?

Mark only one oval.

- ☐ 0-18 years old
☐ 18-30 years old
☐ 30-60 years old
☐ 60+ years old

2. How do you mostly consume audio media?

Tick all that apply.

- ☐ Streaming platforms - Spotify, Apple Music, Yandex Music and others.
☐ Vinyl, CD discs and other physical media.
☐ Downloaded files.
☐ Other: _____

3. How often do you listen to music?

Tick all that apply.

- ☐ Everyday.
☐ Often, but not everyday - e.g. 3-5 days a week.
☐ Once or twice a week at most.
☐ Only on special occasions.
☐ Other: _____

4. Roughly - how much songs do you listen to in a month?

Tick all that apply.

- ☐ 1000+
☐ 500-1000
☐ 100-500
☐ 50-100
☐ 10-50
☐ Less than 10
☐ Other: _____

5. Would you say that you are consistent with the music you listen to? I.e. do you mostly listen to the same artists/songs/genres?

Mark only one oval.

- ☐ Yes
☐ Mostly
☐ Rather no than yes
☐ No

6. How often do you listen to something new?

Mark only one oval.

- ☐ Very rarely, once a couple of months.
☐ Rarely, maybe a handful times a month.
☐ Pretty often, every week or so.
☐ Often, a couple of times a week.
☐ Every or almost every day.

7. How do you find out about new music?

Tick all that apply.

- ☐ At random - in cafes, shops and other venues.
- ☐ From people I know.
- ☐ Non-specialized online platforms - Instagram, TikTok, Youtube(videos), etc.
- ☐ Specialized online platforms - Rate Your Music, Pitchfork, Wire, etc.
- ☐ Games, movies, other forms of digital content.
- ☐ Buying new physical releases(vinyl, cds ...)
- ☐ Recommendations on streaming platforms
- ☐ Other: _____

8. Would you say that you know a lot of people with similar music taste?

Mark only one oval.

- ☐ Yes
- ☐ No

9. Would you like to connect with others who share your musical taste, or influence those around you to explore and appreciate the music you enjoy?

Mark only one oval.

- ☐ Yes
- ☐ No, I already have enough people around me with whom I can share the music I like
- ☐ No, I do not feel the need in that
- ☐ Other: _____

10. Are there any situations, when you listen to music with someone else?

Tick all that apply.

- ☐ At parties; when I'm visiting someone; in a car.
- ☐ Online via Spotify Jam, Discord bots and other similar instruments.
- ☐ Concerts, clubs, street performances.
- ☐ At work.
- ☐ Other: _____

11. How often do you listen to music with someone else?

Mark only one oval.

- ☐ Very rarely, maybe a couple times a year.
- ☐ Rarely, every one or two months.
- ☐ Often, a couple of times a month.
- ☐ Very often, every week or so.
- ☐ Almost every day.
- ☐ Other: _____

12. How often do you/your friends share/discuss music?

Mark only one oval.

- ☐ Very rarely, maybe a couple times a year.
- ☐ Rarely, every one or two months.
- ☐ Often, a couple of times a month.
- ☐ Very often, every week or so.
- ☐ Almost every day.
- ☐ Other: _____

13. How does that happen?

Tick all that apply.

- ☐ Online, sending links from streaming platforms via messengers.
- ☐ Offline, in a car, at someone's place etc.
- ☐ Via sharing the physical media(e.g. giving a vinyl to a friend for a listen)
- ☐ Other: _____

14. Question for users of streaming platforms.

What platform do you use?

Tick all that apply.

- ☐ Spotify
- ☐ Apple Music
- ☐ Yandex Music
- ☐ VK Music
- ☐ SoundCloud
- ☐ Bandcamp
- ☐ Youtube Music
- ☐ Other: _____

15. Question for users of streaming platforms.

Would you benefit, if current streaming platforms added extra social features(i.e. would you use them)? For example, chat discussions for playlists and songs, feeds with music added by the people you follow, forums, direct messages, etc.

Mark only one oval.

- ☐ I would use these features often.
- ☐ I would use these features rarely.
- ☐ I wouldn't use these features.

Music Consumption Study

https://docs.google.com/forms/u/5/d/1vhhAu_SfuHV4xy6JoDaRsM5...

16. Question for users of streaming platforms.

Are there any other features you would like to see?

This content is neither created nor endorsed by Google.

Google Forms

Music Consumption Study

https://docs.google.com/forms/u/5/d/1vhhAu_SfuHV4xy6JoDaRsM5...

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